KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION · 2026

Yearly Test Revision

Everything you need for the Year 10 Mathematics Yearly Test

NAME _____ **CLASS** _____

TEST TIME	TOTAL MARKS	CALCULATORS	BRING
1 hour 15 min	**55 marks**	**Allowed**	**Pen, pencil**
75 minutes	4 sections	bring your own	ruler, eraser, calculator

HOW TO USE THIS BOOKLET

1. Read the **key ideas** and **examples** for each section (pages 2–5).
2. Do the **Try these** questions. Work them out in your book, then write the answer on the line.
3. Check your answers on **page 6**. Tick the checklist below.

MY CHECKLIST

A Financial mathematics — page 2

	Can do	Practise
Percentages	☐	☐
Increase and decrease	☐	☐
GST (10%)	☐	☐
Profit, mark-up, discount	☐	☐
Wages and overtime	☐	☐
Commission, gross and net pay	☐	☐
Loans	☐	☐
Simple and compound interest	☐	☐

B Measurement — page 3

	Can do	Practise
Converting units	☐	☐
Scientific notation	☐	☐
Perimeter	☐	☐
Circles	☐	☐
Area	☐	☐
Surface area	☐	☐
Volume	☐	☐
Capacity	☐	☐

C Statistics and probability — page 4

	Can do	Practise
Types of data	☐	☐
Mean, median, mode, range	☐	☐
Stem-and-leaf plots	☐	☐
Frequency tables	☐	☐
Histograms and shape	☐	☐
Time-series graphs	☐	☐
Probability	☐	☐

D Other Year 10 topics — page 5

	Can do	Practise
Index laws	☐	☐
Substitution	☐	☐
Solving equations	☐	☐
Straight-line graphs	☐	☐
Pythagoras' theorem	☐	☐
Trigonometry	☐	☐

SECTION A

Financial mathematics

In the test: Questions **1–11** · **15 marks** · about **20 minutes**

KEY IDEAS

Percentages

% → decimal: ÷ 100 (15% = 0.15)
Score as a %: score ÷ total × 100

EXAMPLE 15% of $240 = 0.15 × 240 = $36

Increase and decrease

Use a multiplier:
up 8% → × 1.08 · down 8% → × 0.92

EXAMPLE Increase $50 by 10%: 50 × 1.1 = $55

GST (10%)

Add GST: × 1.1
GST in a price (GST inc.): ÷ 11

EXAMPLE $44 (GST inc.): GST = 44 ÷ 11 = $4

Profit, mark-up, discount

Profit = selling price − cost price
Mark-up 20% → × 1.2 · discount 25% (25% off) → × 0.75

EXAMPLE Cost $10, mark-up 20%: 10 × 1.2 = $12, profit = $2

Wages and overtime

Pay = hours × hourly rate
Time and a half × 1.5 · double time × 2

EXAMPLE $20 an hour, 3 hours at time and a half: 3 × 1.5 × 20 = $90

Commission, gross and net pay

Commission = % of sales
Gross pay = total pay before anything is taken out
Net pay (take-home pay) = gross pay − tax − super

EXAMPLE 5% of $2 000 = 0.05 × 2000 = $100

Loans

Total repaid = repayment × number of months (1 year = 12 months, 2 years = 24 months)
Interest = total repaid − amount borrowed

EXAMPLE Borrow $500, pay $50 a month for 1 year: 50 × 12 = $600, interest $100

Simple and compound interest

Simple: $I = P \times R \times N$ · Compound: $A = P(1 + R)^n$
P = amount invested, R = rate as a decimal (5% = 0.05), N or n = number of years

EXAMPLE $1 000 at 5% for 2 years: simple 1000 × 0.05 × 2 = $100; compound $1000 \times 1.05^2 = \$1102.50$, interest $102.50

TRY THESE

Work them out in your book · answers on page 6

1 A shop buys a hat for $12. It adds a 25% mark-up. How much profit does the shop make?

$ _____

2 A jacket costs $66 (GST inc.). How much GST is in the price?

$ _____

3 Shoes cost $90. They are 30% off. Find the new price.

$ _____

4 Lena earns $18 an hour. She works 2 hours of overtime at time and a half. How much is the overtime pay?

$ _____

5 Jack earns 3% commission on sales of $1 500. Find his commission.

$ _____

6 Ava's gross pay is $900. She pays $120 tax and $40 super. Find her net pay.

$ _____

7 Sam borrows $800 and repays $40 a month for 2 years. How much interest does he pay?

$ _____

8 $1 000 is invested at 6% a year for 2 years. Find the interest with simple interest, then with compound interest.

simple $ _____ compound $ _____

SECTION B

Measurement

In the test: Questions **12–21** · **15 marks** · about **20 minutes**

KEY IDEAS

Converting units

km → m × 1000 · m → cm × 100 · cm → mm × 10
L → mL × 1000 · kg → g × 1000 · going back: ÷

EXAMPLE 2.4 km = 2.4 × 1000 = 2 400 m

Scientific notation

One digit (1 to 9) before the decimal point, × a power of 10.
The power = how many places the point moves.

EXAMPLE 3 400 000 = 3.4×10^6 (6 places)
$0.0005 = 5 \times 10^{-4}$ (small: negative power)

Perimeter

Add up the lengths of all the sides.
Rectangle: 2 × (length + width)

EXAMPLE 8 m by 3 m: 2 × (8 + 3) = 22 m

Circles

$C = 2 \times \pi \times r$ (circumference)
$A = \pi \times r \times r$ (area)
r is the radius (half the diameter)



EXAMPLE Radius 5 cm:
$C = 2 \times \pi \times 5 = 31.4$ cm
$A = \pi \times 5 \times 5 = 78.5$ cm^2

Area

Rectangle: length × width
Triangle: ½ × base × height
Trapezium: ½ × (top + bottom) × height (top and bottom are the parallel sides)

EXAMPLE Triangle with base 8 cm, height 5 cm:
½ × 8 × 5 = 20 cm^2

Surface area

Add the area of every face.
A box has 3 pairs of matching faces.

EXAMPLE 4 cm × 3 cm × 2 cm box: faces 12, 8 and 6 cm^2; 2 × (12 + 8 + 6) = 52 cm^2

Volume

Box: length × width × height (cube: side × side × side)
Cylinder: $V = \pi \times r \times r \times h$

EXAMPLE $r = 3$ cm, $h = 10$ cm:
$\pi \times 3 \times 3 \times 10 = 282.7$ cm^3

Capacity

1 cm^3 holds 1 mL
1 m^3 holds 1000 L

EXAMPLE A box with volume 250 cm^3 holds 250 mL

TRY THESE

Work them out in your book · answers on page 6

1 Change 4.2 km into metres, and 3.5 L into millilitres.

_____ m _____ mL

2 Write 6 300 000 in scientific notation.

_____

3 Find the perimeter of a rectangle 9 m long and 5 m wide.

_____ m

4 A circle has radius 4 cm. Find its circumference, to 1 decimal place.

_____ cm

5 A circle has radius 4 cm. Find its area, to 1 decimal place.

_____ cm^2

6 A trapezium has parallel sides 4 cm and 8 cm, and height 5 cm. Find its area.

_____ cm^2

7 Find the surface area of a box 5 cm × 3 cm × 2 cm.

_____ cm^2

8 A cylinder has radius 3 cm and height 5 cm. Find its volume (to 1 d.p.) and how many mL it holds.

_____ cm^3 _____ mL

SECTION C

Statistics and probability

In the test: Questions **22–29** · **14 marks** · about **20 minutes**

KEY IDEAS

Types of data

Categorical (words): nominal, no order (eye colour) or ordinal, in order (small, medium, large)
Numerical (numbers): discrete, counted (number of goals) or continuous, measured (height)

Mean, median, mode, range

Mean: add them up, ÷ how many
Median: middle number (in order)
Mode: most common · Range: biggest − smallest

EXAMPLE 3, 4, 4, 13, 16: mean 40 ÷ 5 = 8, median 4, mode 4, range 16 − 3 = 13

Stem-and-leaf plots

The stem is the tens digit. Each leaf is a ones digit.

EXAMPLE Stem 2, leaf 5 means 25

Stem	Leaf
1	3 8
2	5 5 7

2 | 5 means 25

Frequency tables

Total = add up the frequency column
Percentage frequency = frequency ÷ total × 100

EXAMPLE 6 out of 24: 6 ÷ 24 × 100 = 25%

Histograms and shape

A column graph for groups. Each column is as tall as its frequency. The columns touch.



symmetrical skewed

Time-series graphs

Data over time. The trend is the overall direction.

EXAMPLE The line goes up over time: **upward trend**. The line goes down: **downward trend**.

Probability

Probability = number of ways it can happen ÷ total number of ways
P(red) means "the probability of red" · 0 = impossible · 1 = certain

EXAMPLE A bag has 3 red and 7 blue counters: P(red) = $\frac{3}{10}$

TRY THESE

Work them out in your book · answers on page 6

1 What type of data is "hair colour"?

_____

2 Data: 4, 9, 2, 8, 4, 12, 10. Find the mean.

_____

3 Data: 4, 9, 2, 8, 4, 12, 10. Find the median and the range.

median = _____ range = _____

4 Use the stem-and-leaf plot. Find the lowest score and the mode.

Stem	Leaf
1	2 5
2	0 3 3 8
3	1 4

1 | 2 means 12

lowest = _____ mode = _____

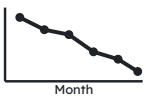
5 5 out of 25 students walk to school. What percentage is this?

_____ %

6 A letter is chosen from COCONUT. Find P(C) and P(vowel). (Vowels: A, E, I, O, U)

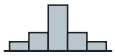
P(C) = _____ P(vowel) = _____

7 Describe the trend of this graph.



_____

8 Is this histogram symmetrical or skewed?



_____

SECTION D

Other Year 10 topics

In the test: Questions **30–35** · **11 marks** · about **15 minutes**

KEY IDEAS

Index laws

Multiply: **add** the powers
Divide: **subtract** the powers

EXAMPLE $y^3 \times y^2 = y^5$ · $y^6 \div y^2 = y^4$

Substitution

Put the number in place of the letter. $3a$ means $3 \times a$.

EXAMPLE $a = 5$: $3a + 2 = 3 \times 5 + 2 = 17$

Solving equations

Do the **opposite** to both sides.

EXAMPLE $2x + 5 = 17$ → take away 5: $2x = 12$ → divide by 2: $x = 6$
$\frac{x+3}{2} = 5$ → times by 2: $x + 3 = 10$ → take away 3: $x = 7$

Straight-line graphs

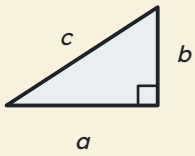
Make a table of values, plot the points, join with a ruler.
$y = mx + c$: m is the gradient (how steep the line is), c is the y-intercept (where the line crosses the y-axis)

EXAMPLE $y = 5x + 2$: gradient 5, y-intercept 2

Pythagoras' theorem

$c^2 = a^2 + b^2$
c is the hypotenuse, the longest side

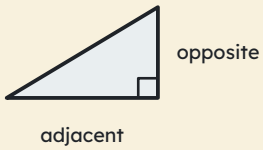
EXAMPLE $a = 5$, $b = 12$:
$c^2 = 25 + 144 = 169$
$c = \sqrt{169} = 13$



Trigonometry

Use tan: tan (angle) = opposite ÷ adjacent
so opposite = adjacent × tan (angle)

EXAMPLE Angle 30°, adjacent 10 cm:
$10 \times \tan 30° = 5.8$ cm
Calculator in **DEG** mode.



TRY THESE

Work them out in your book · answers on page 6

1 Simplify $p^5 \times p^3$.

_____

2 If $b = 6$, find $4b - 5$.

_____

3 Solve $3x + 4 = 22$.

$x =$ _____

4 Solve $\frac{x-1}{2} = 4$.

$x =$ _____

5 For $y = 3x + 2$, find y when $x = 0$, 1 and 2. Fill in the table.

x	0	1	2
y			

6 Write the gradient and y-intercept of $y = 3x + 2$.

gradient = _____ y-intercept = _____

7 A right-angled triangle has short sides 9 cm and 12 cm. Find the hypotenuse.

_____ cm

8 A right-angled triangle has an angle of 40°. The side next to it (adjacent) is 10 cm. Find the opposite side, to 1 decimal place.

_____ cm

CHECK

Answers

The working is shown in grey. Got one wrong? Read that key idea again, then try it once more.

A Financial mathematics

1. **$3.00** — 12 × 1.25 = $15; 15 − 12
2. **$6.00** — 66 ÷ 11
3. **$63.00** — 90 × 0.7
4. **$54.00** — 2 × 1.5 × 18
5. **$45.00** — 0.03 × 1500
6. **$740.00** — 900 − 120 − 40
7. **$160.00** — 40 × 24 = $960; 960 − 800
8. **simple $120.00, compound $123.60** — 1000 × 0.06 × 2; 1000 × 1.06^2 = 1123.60, − 1000

B Measurement

1. **4 200 m, 3 500 mL** — 4.2 × 1000; 3.5 × 1000
2. **6.3×10^6** — 6.3 (one digit before the point); the point moves 6 places
3. **28 m** — 2 × (9 + 5)
4. **25.1 cm** — 2 × π × 4
5. **50.3 cm^2** — π × 4 × 4
6. **30 cm^2** — ½ × (4 + 8) × 5
7. **62 cm^2** — 2 × (15 + 10 + 6)
8. **141.4 cm^3, 141.4 mL** — π × 3 × 3 × 5; 1 cm^3 holds 1 mL

C Statistics and probability

1. **categorical, nominal** — words, no order
2. **7** — 49 ÷ 7
3. **median 8, range 10** — in order: 2, 4, 4, 8, 9, 10, 12; range 12 − 2
4. **lowest 12, mode 23** — 1 | 2 means 12; 23 appears twice
5. **20%** — 5 ÷ 25 × 100
6. **2̲7, 3̲7** — 2 C's out of 7 letters; O, O, U
7. **downward trend** — the line goes down over time
8. **symmetrical** — both sides match

D Other Year 10 topics

1. **p^8** — add the powers: 5 + 3
2. **19** — 4 × 6 − 5
3. **$x = 6$** — take away 4: $3x = 18$; divide by 3
4. **$x = 9$** — times by 2: $x - 1 = 8$; add 1
5. **2, 5, 8** — 3 × 0 + 2, 3 × 1 + 2, 3 × 2 + 2
6. **gradient 3, y-intercept 2** — $m = 3$, $c = 2$
7. **15 cm** — 81 + 144 = 225, √225
8. **8.4 cm** — 10 × tan 40°

ON TEST DAY

- Check your calculator is in **DEG** mode.
- Read each question twice. Look for the **HELP** box.
- Show your working and write the **units**.
- Answer every question. Never leave a blank.

CALCULATOR KEYS

- **^** or **$x^{\blacksquare}$** for powers: 1.05 ^ 2
- **π** for circles · **√** for Pythagoras
- **tan** for trigonometry: 10 × tan 40
- Use brackets: (8 + 3) × 2